

23.1 Resampling cases versus resampling residuals

Let $\theta = \mathbf{c}^\top \boldsymbol{\beta}$ be estimated by $\hat{\theta} = \mathbf{c}^\top \hat{\boldsymbol{\beta}}$. An interval or a test for θ needs the distribution of a **root** such as $\hat{\theta} - \theta$, or a studentized version of it. That distribution depends on the mechanism P that produced the data. In Part III, P lay in the normal linear model, where the studentized root is $t(n-r)$ whatever the parameters. Without that assumption the distribution is unknown.

The **bootstrap** (Efron 1979) replaces P by an estimate $\hat{P}$. Draw a data set $\mathbf{Y}^*$ from $\hat{P}$, compute $\hat{\theta}^*$ from it, and use the distribution of $\hat{\theta}^* - \hat{\theta}$, given the data, as the approximation to that of $\hat{\theta} - \theta$. Here $\hat{\theta}$ plays the true value, because it is the value of θ under $\hat{P}$. Probabilities, expectations and covariances under $\hat{P}$ with the data held fixed are written Pr_* , E_* and Cov_* . In practice B resamples approximate the bootstrap distribution. Their Monte Carlo error shrinks as B grows; the statistical error, from the gap between $\hat{P}$ and P , does not, and it is the subject of this chapter.

23.1.1 Two ways to estimate the data-generating mechanism

In the **fixed-design model** the rows $\mathbf{x}_{(i)}^\top$ of $\mathbf{X}$ are constants and $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with independent, identically distributed errors from an unknown law F . Estimate $\boldsymbol{\beta}$ by $\hat{\boldsymbol{\beta}}$ and F by the empirical distribution of the residuals. In the **random-design model** the cases $(\mathbf{x}_{(i)}, Y_i)$ are independent draws from an unknown joint distribution. Estimate it by the empirical distribution of the n cases.

Definition 23.1.1. Residual and case bootstraps

Let $\mathbf{X}$ be $n \times p$ of rank p , with least squares estimate $\hat{\boldsymbol{\beta}}$ and residuals $\hat{\varepsilon}_1, \dots, \hat{\varepsilon}_n$ with mean $\hat{\bar{\varepsilon}}$.

- (a) The **residual bootstrap** draws $\varepsilon_1^*, \dots, \varepsilon_n^*$ independently and uniformly from the centred residuals $\tilde{\varepsilon}_i = \hat{\varepsilon}_i - \hat{\bar{\varepsilon}}$, sets $\mathbf{Y}^* = \mathbf{X}\hat{\boldsymbol{\beta}} + \boldsymbol{\varepsilon}^*$, and computes $\hat{\boldsymbol{\beta}}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}^*$. The design is the same in every resample.
- (b) The **case bootstrap** (or pairs bootstrap) draws n indices $I_1, \dots, I_n$ independently and uniformly from $\{1, \dots, n\}$ and computes the least squares estimate $\hat{\boldsymbol{\beta}}^*$ from the cases $(\mathbf{x}_{(I_k)}, Y_{I_k})$, $k = 1, \dots, n$. The design changes from resample to resample.

Variants of (a) resample the **leverage-adjusted residuals** $r_i = \hat{\varepsilon}_i / \sqrt{1 - h_{ii}}$, centred, or the residuals rescaled by $\sqrt{n/(n-p)}$.

With an intercept the residuals already sum to zero (Proposition 6.4.3(b)). Without one, centring prevents a bias in the bootstrap world ([Exercise \(A1\)](#)).

23.1.2 The moments of the residual bootstrap

Proposition 23.1.1. Moments of the residual bootstrap

In the residual bootstrap, put $\hat{\sigma}_*^2 = n^{-1} \sum_i \tilde{\varepsilon}_i^2$, the variance of the resampled law. Then:

- (a) $E_*(\hat{\boldsymbol{\beta}}^*) = \hat{\boldsymbol{\beta}}$ and $\text{Cov}_*(\hat{\boldsymbol{\beta}}^*) = \hat{\sigma}_*^2(\mathbf{X}^\top \mathbf{X})^{-1}$;
- (b) if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, then $\hat{\sigma}_*^2 = \text{SSE}/n = (n-p)s^2/n$;
- (c) if the data follow the linear model with $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, then $E(r_i^2) = \sigma^2$ for each i with $h_{ii} < 1$.

Proof.

- (a) Given the data, $\hat{\boldsymbol{\beta}}^* = \hat{\boldsymbol{\beta}} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}^*$, where the ε_i^* are independent with $E_*(\varepsilon_i^*) = n^{-1} \sum_j \tilde{\varepsilon}_j = 0$ and $\text{Var}_*(\varepsilon_i^*) = \hat{\sigma}_*^2$. Apply Theorem 5.5.1 in the bootstrap world. (b) The residuals sum to zero, so $\tilde{\varepsilon}_i = \hat{\varepsilon}_i$. (c) $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ by Proposition 6.4.3(e), so $E(\hat{\varepsilon}_i) = 0$ and $\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$. ■

By (b) the plain residual bootstrap shrinks the usual variance by the factor $(n-p)/n$: residuals are shorter than errors because least squares fits part of the noise (Theorem 5.6.1). The factor matters only when p/n is not small (Section 23.6). Rescaling by $\sqrt{n/(n-p)}$ removes it exactly when the model has an intercept. The leverage-adjusted residuals remove it observation by observation. They are s times the internally studentized residuals of Chapter 20 (Definition 20.1.1).

23.1.3 Consistency for a linear contrast

Consider designs $\mathbf{X}_n$ of rank p , with p fixed, and errors $\varepsilon_{n1}, \dots, \varepsilon_{nn}$ drawn independently from a fixed law F with mean zero and variance $\sigma^2 \in (0, \infty)$. For $\mathbf{c}_n \neq \mathbf{0}$ put

$$\begin{aligned} \mathbf{a}_n &= \mathbf{X}_n(\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{c}_n, & \mathbf{c}_n^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) &= \mathbf{a}_n^\top \boldsymbol{\varepsilon}, \\ \|\mathbf{a}_n\|^2 &= \mathbf{c}_n^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{c}_n, \end{aligned}$$

so the contrast estimate is a weighted sum of the errors with standard deviation $\sigma \|\mathbf{a}_n\|$. The condition we need is that no single weight dominates:

$$m_n = \max_i \frac{|a_{ni}|}{\|\mathbf{a}_n\|} \rightarrow 0. \quad (23.1.1)$$

The Cauchy–Schwarz bound $a_{ni}^2 \leq h_{ii} \|\mathbf{a}_n\|^2$ in the proof of Theorem 19.3.2 shows that Eq. 23.1.1 holds for every contrast when the largest leverage tends to zero. The limit theorem used there is the Lindeberg–Feller theorem of Chapter 19; for a triangular array U_{ni} its condition holds in particular when $\sum_i E U_{ni}^4 \rightarrow 0$ (Lyapunov), since the Lindeberg sum is at most $\eta^{-2} \sum_i E U_{ni}^4$. We add one more result from probability, used without proof (van der Vaart 1998, chapter 2).

Remark (Pólya’s theorem). If distribution functions G_n converge pointwise to a continuous distribution function G , then $\sup_x |G_n(x) - G(x)| \rightarrow 0$.

A bootstrap distribution is a conditional distribution that changes with the data, so we need a conditional form of the Lindeberg–Feller theorem.

Lemma 23.1.2. A conditional central limit theorem

For each n , given the data $\mathcal{D}_n$, let $U_{n1}^*, \dots, U_{nn}^*$ be independent with $E_*(U_{ni}^*) = 0$ and $\sum_i E_*(U_{ni}^{*2}) = 1$. If for every $\eta > 0$

$$L_n^*(\eta) = \sum_i E_*[U_{ni}^{*2} \mathbf{1}\{|U_{ni}^*| > \eta\}] \rightarrow 0 \quad \text{in probability,}$$

then $\Delta_n = \sup_x |\Pr_*(\sum_i U_{ni}^* \leq x) - \Phi(x)| \rightarrow 0$ in probability.

Proof. Random variables converge to zero in probability if and only if every subsequence has a further subsequence along which they converge almost surely. Given a subsequence, a diagonal argument gives a further subsequence along which $L_n^*(1/k) \rightarrow 0$ almost surely for every $k = 1, 2, \dots$ at once. Since $L_n^*(\eta)$ decreases in η , on this event $L_n^*(\eta) \rightarrow 0$ for every $\eta > 0$. For each outcome in the event the conditional laws satisfy the Lindeberg condition, so $\Delta_n \rightarrow 0$ by the Lindeberg–Feller and Pólya theorems. ■

Lemma 23.1.3. Residuals have uniformly small tails

In the setting above, $\hat{\sigma}_*^2 \rightarrow \sigma^2$ in probability, and for every sequence of constants $K_n \rightarrow \infty$, $n^{-1} \sum_i \tilde{\varepsilon}_i^2 \mathbf{1}\{|\tilde{\varepsilon}_i| > K_n\} \rightarrow 0$ in probability.

Proof. Write $\tilde{\varepsilon}_i = \varepsilon_i - \delta_i$ with $\boldsymbol{\delta} = \mathbf{M}\boldsymbol{\varepsilon} + \bar{\varepsilon}\mathbf{1}$. Here $E\|\mathbf{M}\boldsymbol{\varepsilon}\|^2 = \sigma^2 p$, and $\bar{\varepsilon} = n^{-1}\mathbf{1}^\top(\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ has $E(\bar{\varepsilon}^2) = \sigma^2 n^{-2}\mathbf{1}^\top(\mathbf{I} - \mathbf{M})\mathbf{1} \leq \sigma^2/n$. So $E\|\boldsymbol{\delta}\|^2 \leq 2\sigma^2(p+1)$, and $n^{-1}\|\boldsymbol{\delta}\|^2 \rightarrow 0$ in probability.

First claim: $\hat{\sigma}_*^2 = n^{-1}\|\boldsymbol{\varepsilon}\|^2 - n^{-1}\|\mathbf{M}\boldsymbol{\varepsilon}\|^2 - \bar{\varepsilon}^2$. The first term tends to σ^2 by the weak law of large numbers (the errors of the n th model are an independent sample from F), and the others tend to zero.

Second claim: for $K > 0$ and reals u, d , with $v = u - d$,

$$v^2 \mathbf{1}\{|v| > K\} \leq 2u^2 \mathbf{1}\{|u| > K/2\} + 4d^2.$$

If $|u| > K/2$, use $v^2 \leq 2u^2 + 2d^2$. If $|u| \leq K/2$ and $|v| > K$, then $|d| \geq |v| - |u| > K/2 \geq |u|$, so $|v| \leq 2|d|$. Otherwise the left side is zero. With $u = \varepsilon_i$, $d = \delta_i$ and $K = K_n$, averaging gives

$$\begin{aligned} \frac{1}{n} \sum_i \tilde{\varepsilon}_i^2 \mathbf{1}\{|\tilde{\varepsilon}_i| > K_n\} &\leq \frac{2}{n} \sum_i \varepsilon_i^2 \mathbf{1}\{|\varepsilon_i| > K_n/2\} \\ &\quad + \frac{4}{n} \|\boldsymbol{\delta}\|^2. \end{aligned}$$

The first term has expectation $2E[\varepsilon^2 \mathbf{1}\{|\varepsilon| > K_n/2\}] \rightarrow 0$ by dominated convergence, and the second tends to zero in probability. ■

Theorem 23.1.4. Consistency of the residual bootstrap

In the fixed-design setting above, assume [Eq. 23.1.1](#). Let

$$G_n(x) = \Pr(\mathbf{c}_n^\top(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \leq x \|\mathbf{a}_n\|),$$

$$\hat{G}_n(x) = \Pr_*(\mathbf{c}_n^\top(\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}}) \leq x \|\mathbf{a}_n\|),$$

where $\hat{\boldsymbol{\beta}}^*$ comes from the residual bootstrap. Then:

- (a) $\sup_x |G_n(x) - \Phi(x/\sigma)| \rightarrow 0$;
- (b) $\sup_x |\hat{G}_n(x) - \Phi(x/\sigma)| \rightarrow 0$ in probability;
- (c) $\sup_x |\hat{G}_n(x) - G_n(x)| \rightarrow 0$ in probability.

Proof. Put $w_i = a_{ni}/\|\mathbf{a}_n\|$, so $\sum_i w_i^2 = 1$ and $\max_i |w_i| = m_n \rightarrow 0$.

- (a) $\mathbf{c}_n^\top(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\|\mathbf{a}_n\| = \sum_i w_i \varepsilon_i$. When $\max_i h_{ii} \rightarrow 0$ the claim is Theorem 19.3.2(a) with Pólya's theorem. The same Lindeberg argument needs only [Eq. 23.1.1](#): with $U_{ni} = w_i \varepsilon_i / \sigma$, the Lindeberg sum is at most $E[\sigma^{-2} \varepsilon_i^2 \mathbf{1}\{|\varepsilon_i| > \eta \sigma / m_n\}] \rightarrow 0$, by dominated convergence.
- (b) Similarly $\mathbf{c}_n^\top(\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}})/\|\mathbf{a}_n\| = \sum_i w_i \varepsilon_i^*$. Take $U_{ni}^* = w_i \varepsilon_i^* / \hat{\sigma}_*$. As in (a),

$$L_n^*(\eta) \leq \frac{1}{n \hat{\sigma}_*^2} \sum_i \varepsilon_i^2 \mathbf{1}\{|\varepsilon_i| > \eta \hat{\sigma}_* / m_n\}.$$

By [Lemma 23.1.3](#), $\hat{\sigma}_* \rightarrow \sigma$ in probability. On the event $\hat{\sigma}_* > \sigma/2$, whose probability tends to one, the right side is at most $4\sigma^{-2} n^{-1} \sum_i \varepsilon_i^2 \mathbf{1}\{|\varepsilon_i| > K_n\}$ with $K_n = \eta \sigma / (2m_n) \rightarrow \infty$, which tends to zero in probability by the lemma. So [Lemma 23.1.2](#) gives $\sup_x |\hat{G}_n(x) - \Phi(x/\hat{\sigma}_*)| \rightarrow 0$ in probability. Finally $\sup_x |\Phi(x/\hat{\sigma}_*) - \Phi(x/\sigma)|$ is a continuous function of $\hat{\sigma}_*/\sigma$ that vanishes at 1, so it tends to zero in probability.

- (c) follows from (a) and (b) by the triangle inequality. ■

The theorem holds for the rescaled residuals too, since they differ by the factor $\sqrt{n/(n-p)} \rightarrow 1$; for the leverage-adjusted ones the proof adapts when $\max_i h_{ii} \rightarrow 0$, with small changes we omit. It needs neither normal errors nor moments beyond the second. Freedman (1981) proved it in a stronger metric, together with a statement for the case bootstrap. It does not say that the bootstrap beats the normal approximation $\Phi(x/s)$, which is consistent under the same conditions. The bootstrap earns its keep through studentization ([Section 23.3](#)) and through its flexibility for statistics with no convenient standard error.

Warning. The residual bootstrap treats the residuals as exchangeable. It gives $\text{Cov}_*(\hat{\boldsymbol{\beta}}^*) = \hat{\sigma}_*^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ whatever the data look like, which is the classical formula, so it inherits that formula's failure under heteroscedasticity (Theorem 19.2.2, 1). Wu (1986) made this point forcefully.

23.1.4 The case bootstrap

The case bootstrap assumes only that the cases are independent draws from a population. By Proposition 6.11.3, least squares then estimates the population projection coefficient whether or not the regression function is linear, and the case bootstrap estimates the variability of $\hat{\beta}$ around that target.

Let W_i be the number of times case i is drawn, so that $(W_1, \dots, W_n)$ is multinomial with n trials and equal probabilities, and put $\mathbf{W} = \text{diag}(W_i)$. When $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ is invertible, $\hat{\beta}^* = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}$, and since $\mathbf{y} = \mathbf{X} \hat{\beta} + \hat{\epsilon}$,

$$\hat{\beta}^* - \hat{\beta} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \hat{\epsilon}. \quad (23.1.2)$$

The matrix $n^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$ has expectation $n^{-1} \mathbf{X}^\top \mathbf{X}$ and is close to it for large n . Replacing it by its expectation gives a linear approximation whose covariance is exact.

Proposition 23.1.5. The case bootstrap, linearized, is the sandwich

Let $\mathbf{L}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \hat{\epsilon}$. Then $E_*(\mathbf{L}^*) = \mathbf{0}$ and

$$\text{Cov}_*(\mathbf{L}^*) = (\mathbf{X}^\top \mathbf{X})^{-1} \left(\sum_i \hat{\epsilon}_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1},$$

the sandwich covariance estimator HC0 of Chapter 21 (Theorem 21.4.2).

Proof. For multinomial counts with n trials and probabilities $1/n$, $E(W_i) = 1$, $\text{Var}(W_i) = 1 - 1/n$ and $\text{Cov}(W_i, W_j) = -1/n$ for $i \neq j$. So $E_*(\mathbf{W} \hat{\epsilon}) = \hat{\epsilon}$ and $\text{Cov}_*(\mathbf{W} \hat{\epsilon}) = \text{diag}(\hat{\epsilon})(\mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top) \text{diag}(\hat{\epsilon})$. Hence $E_*(\mathbf{L}^*) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \hat{\epsilon} = \mathbf{0}$ by Proposition 6.4.3(a), and $\text{Cov}_*(\mathbf{X}^\top \mathbf{W} \hat{\epsilon}) = \sum_i \hat{\epsilon}_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top - n^{-1} (\mathbf{X}^\top \hat{\epsilon})(\mathbf{X}^\top \hat{\epsilon})^\top = \sum_i \hat{\epsilon}_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$. ■

To first order, then, the case bootstrap estimates the variance of $\hat{\beta}$ by the sandwich, not the classical formula. The price is a varying design: a case of high leverage appears in some resamples several times and in others not at all, and with indicator columns some resamples are singular (Exercise (B2)).

Example. Engel's food expenditure

Return to Engel's data on the income and food expenditure of 235 households (Example (Engel's food expenditure data)). The least squares slope is 0.4852, and the scatter fans out (Figure 23.1.1(a)): the residual standard deviation is 2.36 times larger for the richer half of the households.

The classical standard error of the slope is 0.01437. The residual bootstrap with $B = 4000$ resamples gives 0.01433, near its exact value 0.01431, the classical value times $\sqrt{(n-2)/n} = 0.9957$ (Proposition 23.1.1); with leverage-adjusted residuals the exact value is 0.01484. The case bootstrap gives 0.05032, more than three times as large and close to the HC0 value 0.05177 of Proposition 23.1.5; Chapter 21 computed this and the other sandwich standard errors for the same slope. The residual bootstrap's answer rests on a constant variance that

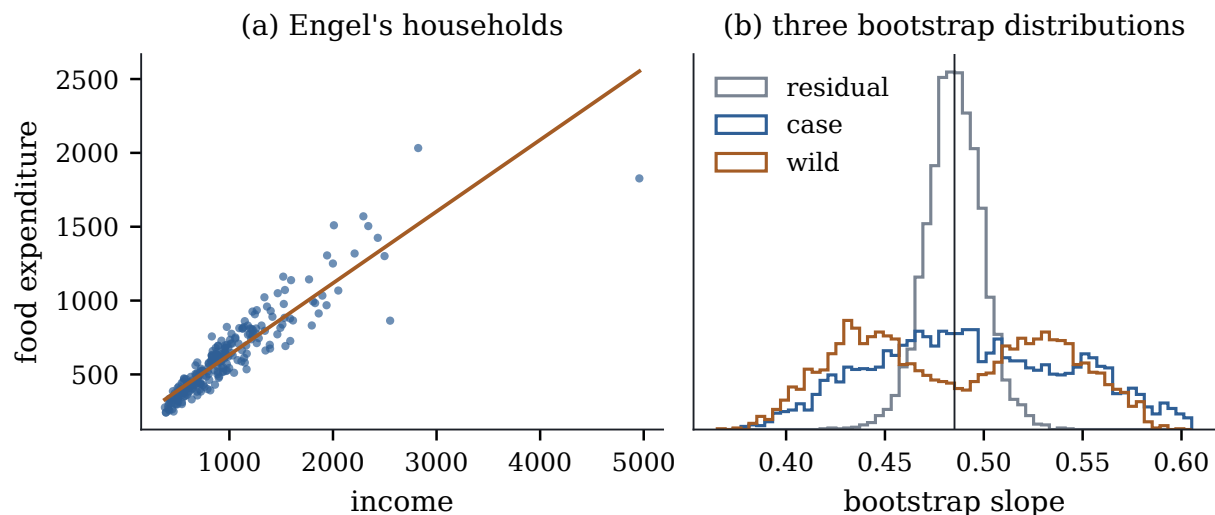


Figure 23.1.1: (a) Engel's households, with the least squares line. (b) Bootstrap distributions of the slope: residual (grey), case (blue) and wild (orange) resampling.

the data contradict.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.engel.load_pandas().data

y = data["foodexp"].to_numpy()
x = data["income"].to_numpy()
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape
XtX_inv = np.linalg.inv(X.T @ X)
beta_hat = XtX_inv @ X.T @ y
e_hat = y - X @ beta_hat
h = np.sum((X @ XtX_inv) * X, axis=1) # leverages
s2 = e_hat @ e_hat / (n - p)
se_classical = np.sqrt(s2 * XtX_inv[1, 1])
print(f"slope {beta_hat[1]:.4f}, classical se {se_classical:.5f}, max leverage {h.
      max():.3f}")

rng = np.random.default_rng(2301)
B = 4000
A = XtX_inv @ X.T # beta_hat = A y
fit = X @ beta_hat
e_c = e_hat - e_hat.mean() # centred residuals
r = e_hat / np.sqrt(1 - h) # leverage-adjusted residuals
r_c = r - r.mean()

def residual_bootstrap(resid, B):
    """Slopes from  $y^* = X \beta_{\text{hat}} + e^*$ , with  $e^*$  drawn with replacement from resid.
    """
    idx = rng.integers(0, n, size=(B, n))
    y_star = fit + resid[idx]
    return (y_star @ A.T)[:, 1]

slopes_res = residual_bootstrap(e_c, B)
slopes_mod = residual_bootstrap(r_c, B)
print(f"residual bootstrap se: raw {slopes_res.std():.5f}, leverage-adjusted {
```

```

slopes_mod.std():.5f}")

def pairs_bootstrap(B):
    """Slopes refitted to n cases drawn with replacement from the (x_i, y_i)."""
    idx = rng.integers(0, n, size=(B, n))
    xs, ys = x[idx], y[idx]
    xc = xs - xs.mean(axis=1, keepdims=True)
    return np.sum(xc * ys, axis=1) / np.sum(xc ** 2, axis=1)

slopes_pairs = pairs_bootstrap(B)
print(f"case bootstrap se {slopes_pairs.std():.5f}")

```

[Run this code online](#)

The choice between the two is a choice of model. The residual bootstrap suits a linear mean with exchangeable errors, even for sampled cases: when the error law does not depend on $\mathbf{X}$, conditioning on the design is justified as in [Section 14.1](#), where Theorem 14.1.2 makes it exact under normality. It is then the more accurate, since it adds no noise from a varying design. Under heteroscedasticity or a nonlinear mean the case bootstrap is safer. The wild bootstrap of the next section keeps the design fixed and still respects heteroscedasticity.

23.1.5 Exercises

23.1.5.1 A. Check your understanding

Exercise. A1

In a regression through the origin, $E(Y_i) = \beta x_i$, the residuals need not sum to zero. Show that if the residual bootstrap resampled the uncentred residuals, then $E_*(\hat{\beta}^*) - \hat{\beta} = \bar{\varepsilon} \sum_i x_i / \sum_i x_i^2$.

Solution. Here $\hat{\beta}^* - \hat{\beta} = \sum_i x_i \varepsilon_i^* / \sum_i x_i^2$, and each uncentred draw has $E_*(\varepsilon_i^*) = n^{-1} \sum_j \varepsilon_j = \bar{\varepsilon}$. Taking expectations gives the stated bias, which is not zero unless $\bar{\varepsilon} = 0$ or $\sum_i x_i = 0$.

Exercise. A2

A case-bootstrap resample of size n contains each original case with probability $1 - (1 - 1/n)^n$. Show that the expected number of distinct cases in a resample is $n(1 - (1 - 1/n)^n)$, about $0.632n$ for large n .

23.1.5.2 B. Practice

Exercise. B1

Suppose the errors are independent with variances σ_i^2 , and the model has an intercept. Show that the residual bootstrap variance of $\mathbf{c}^\top \hat{\boldsymbol{\beta}}^*$ has expectation $\|\mathbf{a}\|^2 n^{-1} \sum_i (1 - h_{ii}) \sigma_i^2$, whereas $\text{Var}(\mathbf{c}^\top \hat{\boldsymbol{\beta}}) = \sum_i a_i^2 \sigma_i^2$. Construct a two-group example in which the ratio of the two is as small as you like.

Solution. By [Proposition 23.1.1](#) the bootstrap variance is $\|\mathbf{a}\|^2 \text{SSE}/n$, and $E(\text{SSE}) = \text{tr}((\mathbf{I} - \mathbf{M})\mathbf{\Sigma}) = \sum_i (1 - h_{ii})\sigma_i^2$ for diagonal $\mathbf{\Sigma}$. The true variance is $\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}$. Take two groups of sizes n_1 and n_2 and the contrast of the group means, so $a_i = 1/n_1$ in group 1 and $-1/n_2$ in group 2. With $\sigma_i^2 = \sigma_1^2$ in the small group and 0 in the large one, the true variance is σ_1^2/n_1 and the expected bootstrap variance is $(1/n_1 + 1/n_2)(n_1 - 1)\sigma_1^2/n$. With n_1 fixed the ratio of the second to the first tends to 0 as $n_2 \rightarrow \infty$.

Exercise. B2

In a one-way layout with g groups of m observations each, find the probability that a case-bootstrap resample omits some group entirely, so that the cell-means model cannot be fitted. Evaluate it for $g = 5$, $m = 4$, and describe a resampling scheme that avoids the problem.

23.1.5.3 C. Going deeper

Exercise. C1

[Exercise \(B2\)](#) shows that a case-bootstrap resample can be one that no least squares fit exists for, when $\mathbf{X}$ holds indicators. The difficulty is not confined to indicators. Take simple regression with an intercept and n distinct values $x_1, \dots, x_n$.

- Show that a resample gives a design of rank 2 unless all n draws pick the same case, an event of probability n^{1-n} . Evaluate it for $n = 5$.
- Deduce that $\text{Cov}_*(\hat{\boldsymbol{\beta}}^*)$ does not exist as written, and say what a program that reports a case-bootstrap standard error must therefore be computing. Contrast $\mathbf{L}^*$ of [Proposition 23.1.5](#).
- Show that a resample which uses exactly two of the cases, i and j , has slope exactly $(y_j - y_i)/(x_j - x_i)$, whatever the multiplicities. Show that the probability of such a resample is $\binom{n}{2}(2^n - 2)/n^n$, and evaluate it for $n = 5$.
- What does (c) say about the tails of the case-bootstrap distribution of the slope, and which feature of the design controls them? Suggest two repairs.

Solution.

- The resampled design has rank 2 as soon as two of the drawn cases have different x values, and the x_i are distinct, so rank 1 requires all n draws to fall on one case. There are n such ordered draws out of n^n , so the probability is n^{1-n} ; for $n = 5$ it is $5^{-4} = 1/625 = 0.0016$.
- On that event $\hat{\boldsymbol{\beta}}^*$ is not defined, so neither is its conditional expectation or covariance: the bootstrap distribution has an atom at “no estimate”. What is reported in practice is the empirical variance over the resamples that could be fitted, that is, the conditional covariance given non-degeneracy, which is a different functional and depends on how failures are handled. The linearization $\mathbf{L}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \hat{\boldsymbol{\epsilon}}$ has no such trouble: it is a fixed linear function of the multinomial counts, and [Proposition 23.1.5](#) computes its covariance exactly.
- If the resample contains only cases i and j , both present, the fitted line passes through (x_i, y_i) and (x_j, y_j) exactly, because a line through two points achieves zero residual sum of squares whatever the multiplicities $a, b \geq 1$ of the two. Its slope is $(y_j - y_i)/(x_j - x_i)$.

For a given pair there are $2^n - 2$ ordered draws using both cases and nothing else, and there are $\binom{n}{2}$ pairs, so the probability is $\binom{n}{2}(2^n - 2)/n^n$; for $n = 5$ it is $10 \cdot 30/3125 = 0.096$.

- (d) Nearly one resample in ten, for $n = 5$, returns the slope of a single pair of points, and a pair with $x_j - x_i$ small can return an enormous one. So the case-bootstrap distribution has tails governed by the smallest gaps between design points, not by the noise; its variance is dominated by rare resamples, and quantile-based intervals are far more stable than the bootstrap standard error. Two repairs: use the linearized form of [Proposition 23.1.5](#), that is the sandwich, or keep the design fixed and resample residuals, in the wild form of [Section 23.2](#) if heteroscedasticity is the worry.

Exercise. C2

Heteroscedasticity is not the only way the residual bootstrap can aim at the wrong variance; a curved mean does it too, with homoscedastic errors. Let X be uniform on $\{-1, 0, 1\}$, let ε be independent of X with mean 0 and variance σ^2 , and let $Y = X^2 + \varepsilon$. A line is fitted, so $\mathbf{x} = (1, X)^\top$.

- Show that the population projection is $\boldsymbol{\beta}^* = (2/3, 0)^\top$, so $U = X^2 - \frac{2}{3} + \varepsilon$, and that $\boldsymbol{\Sigma} = E(\mathbf{x}\mathbf{x}^\top) = \text{diag}(1, \frac{2}{3})$.
- Show that $\boldsymbol{\Omega} = E(U^2\mathbf{x}\mathbf{x}^\top) = \text{diag}(\frac{2}{9} + \sigma^2, \frac{2}{27} + \frac{2}{3}\sigma^2)$, and deduce that the slope entries of $\boldsymbol{\Sigma}^{-1}\boldsymbol{\Omega}\boldsymbol{\Sigma}^{-1}$ and of $E(U^2)\boldsymbol{\Sigma}^{-1}$ are $\frac{1}{6} + \frac{3}{2}\sigma^2$ and $\frac{1}{3} + \frac{3}{2}\sigma^2$.
- Using Theorem 19.5.2 and [Proposition 23.1.1\(a\)](#), identify which of the two numbers the residual bootstrap estimates and which the case bootstrap estimates, and give the ratio of the two standard errors when $\sigma^2 = 0$.
- Here the residual bootstrap is too *wide*, the opposite of
 - Explain why, from where the misfit sits in the design.

Solution. Throughout, $E X = E X^3 = 0$ and $E X^2 = E X^4 = 2/3$, so $\text{Var}(X^2) = 2/3 - 4/9 = 2/9$.

- $\text{Cov}(X, Y) = \text{Cov}(X, X^2) = E X^3 - E X E X^2 = 0$, so the slope of the population projection is 0 and its intercept is $E Y = 2/3$. Then $U = Y - 2/3 = X^2 - 2/3 + \varepsilon$, and $\boldsymbol{\Sigma} = E(\mathbf{x}\mathbf{x}^\top)$ has entries 1, $E X = 0$ and $E X^2 = 2/3$.
- $E U^2 = \text{Var}(X^2) + \sigma^2 = 2/9 + \sigma^2$. Next $E(U^2 X) = E\{(X^2 - 2/3)^2 X\} + 2E\{(X^2 - 2/3)X\}E\varepsilon + \sigma^2 E X = 0$, the first term because $(x^2 - 2/3)^2 x$ is odd and X is symmetric. Finally $E(U^2 X^2) = E\{(X^2 - 2/3)^2 X^2\} + \sigma^2 E X^2$; the first expectation is $\frac{2}{3} \cdot (1 - \frac{2}{3})^2 = 2/27$, since $X^2 = 1$ with probability $2/3$ and $X^2 = 0$ otherwise. With $\boldsymbol{\Sigma}^{-1} = \text{diag}(1, 3/2)$, the slope entry of $\boldsymbol{\Sigma}^{-1}\boldsymbol{\Omega}\boldsymbol{\Sigma}^{-1}$ is $(3/2)^2(2/27 + 2\sigma^2/3) = 1/6 + 3\sigma^2/2$, and that of $E(U^2)\boldsymbol{\Sigma}^{-1}$ is $(2/9 + \sigma^2)(3/2) = 1/3 + 3\sigma^2/2$.
- By Theorem 19.5.2(a) the true limit of $n \text{Var}(\hat{\beta}_1)$ is the sandwich entry $1/6 + 3\sigma^2/2$. The residual bootstrap gives $\text{Cov}_*(\hat{\boldsymbol{\beta}}^*) = \hat{\sigma}_*^2(\mathbf{X}^\top \mathbf{X})^{-1}$ ([Proposition 23.1.1\(a\)](#)), and $n\hat{\sigma}_*^2(\mathbf{X}^\top \mathbf{X})^{-1} \rightarrow E(U^2)\boldsymbol{\Sigma}^{-1}$ by Theorem 19.5.2(b): it estimates $1/3 + 3\sigma^2/2$, the wrong number. The case bootstrap, to first order, is the sandwich HC0 ([Proposition 23.1.5](#)), which estimates the right one (Theorem 21.4.2). With $\sigma^2 = 0$ the residual bootstrap standard error is $\sqrt{(1/3)/(1/6)} = \sqrt{2}$ times too large.
- The residual bootstrap spreads the observed misfit evenly over the design, because it treats the residuals as exchangeable. Here the misfit is not even: the approximation

error $X^2 - 2/3$ is $-2/3$ at $X = 0$ and only $1/3$ at $X = \pm 1$, so the large errors sit at the centre of the design, where they hardly move a slope. Reallocating them to the ends, as resampling does, inflates the slope's variance. Had the misfit been largest at the ends, the same argument would make the residual bootstrap too narrow, as in 1.